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Research Profile

Graduate researcher with a strong foundation in machine learning methods, statistical learning theory, and data-driven decision support. Background includes fairness-aware ML for high-stakes settings, reproducible evaluation frameworks, bandit and sequential decision-making algorithms, and applied ML across healthcare, bioinformatics, and network routing domains. Current Graduate Assistant work (Quantum and AI Research, RIT) emphasizes principled model evaluation, statistical rigor, and method reliability across heterogeneous and partially observed data environments. Interested in pushing ML methodology forward — building models that generalize robustly and whose failure modes are understood and bounded.

Research Interests

Machine learning theory and methods; statistical learning and model evaluation; fairness-aware and robust ML; bandit and sequential decision-making; reproducible benchmarking; applied ML in healthcare, bioinformatics, and network systems.

Education

Rochester Institute of Technology

Expected Aug 2026

Master of Science, Data Science — GPA 3.9

Rochester, NY

- Concentration: Bioinformatics & Artificial Intelligence.
- Relevant coursework: Data-Driven Knowledge Discovery (ISTE-780), Neural Networks & Deep Learning, Applied Data Science (DSCI-601), Bioinformatics Algorithms (BIOL-630), Software Engineering for Data Science (DSCI-644), Foundations of Data Science (DSCI-633).

University of Rochester, Warner School of Education

In progress

Graduate study, Teaching Computer Science K-12 (M.S. track) — GPA 4.0

Rochester, NY

- NSF Robert Noyce Scholar (\$30,000 full funding); Advanced Certificate in Leadership in Disability and Inclusive Practices.

Rochester Institute of Technology

2015

Master of Science, Computer Science — GPA 3.2

Rochester, NY

- Concentrations: Artificial Intelligence, Big Data Analytics, Computer Graphics.
- Graduate research: *Big Data Medical Diagnosis: A Multi-Method Approach* — ML ensemble methods for clinical classification.

Farmingdale State College

2012

Bachelor of Science, Computer Engineering — GPA 3.3

Farmingdale, NY

- Dual degree: Electrical Engineering & Computer Engineering Technology.

- Honors: CSTEP Award & Merit; Dual BS Scholarship.

Research and Professional Experience

Rochester Institute of Technology, Software Engineering Department Fall 2025 – Present
Graduate Assistant, Quantum and AI Research *Rochester, NY*

- Build reproducible ML evaluation pipelines across local, Colab, and GCP environments with structured logging, hyperparameter tracking, and statistical validation.
- Develop bandit and sequential decision-making systems (UCB, neural-UCB, contextual bandits) with rigorous evaluation frameworks comparing algorithmic variants under controlled conditions.
- Produce manuscript-ready technical analyses on ML method performance, including evaluation under distribution shift, partial observability, and adversarial settings.
- Design fairness-aware evaluation protocols that measure both predictive performance and equity metrics across heterogeneous subpopulations.

University of Rochester, Warner School of Education Sep 2024 – Present
CS Teacher Candidate & Education Research (NSF Noyce Scholar) *Rochester, NY*

- Deliver K–12 computer science instruction in active public school placement as part of the NSF Robert Noyce Scholar program, including computational thinking, inclusive pedagogy, and NYS learning standards.
- Conduct applied education research on AI-informed and data-driven approaches to improve learning outcomes for neurodivergent students, applying Universal Design for Learning (UDL) and evidence-based adaptive strategies.
- Produce lesson artifacts and instructional analyses documenting the intersection of AI, inclusive education, and evidence-based CS pedagogy.

Rochester Institute of Technology 2024 – 2025
Equitable ML Research — ISTE-780 Project (Data-Driven Knowledge Discovery) *Rochester, NY*

- Pioneered first systematic application of fairness assessment to RNA secondary structure prediction algorithms (EQUITAS framework), revealing critical equity violations (11.1% equity score) across demographic proxies.
- Developed modular ML evaluation codebase (900+ lines) implementing reproducible fairness-aware algorithm comparison with automated equity-metric reporting.
- Achieved statistical significance ($p < 0.01$) in bias detection across multiple algorithmic variants using SHAP and disparity-impact analysis.

Rochester Institute of Technology 2025 – 2026
Computational Biology Research (BIO614 and BIO550) *Rochester, NY*

- Applied ML and statistical methods to RNA structure prediction, including evaluation of probabilistic models against classical dynamic-programming baselines.
- Conducted RNA-seq differential-expression analysis with reproducible preprocessing, QC validation, and statistical-significance reporting.

VEDADATA Inc. Sep 2022 – Aug 2024
Data Solutions Engineer *New York, NY*

- Designed scalable Python and pandas ML workflows for ingestion, feature engineering, and statistical diagnostics in production environments.
- Implemented structured quality checks and anomaly-detection pipelines strengthening production data reliability.

VIOME Inc.

AI Data Solutions Engineer

Jan 2021 – Sep 2022

New York, NY

- Developed healthcare ML workflows for feature engineering, model experimentation, and evaluation — predictive health applications in AWS production environments.
- Built reproducible model-evaluation infrastructure with monitoring components to detect performance degradation under distribution shift.

Selected Technical Writing

- *Quantum Path Optimization with Fault-Tolerant Routing* — manuscript-level analysis of quantum-classical hybrid ML for fault-tolerant routing under noise constraints (under review, 2025–2026).
- *Big Data Medical Diagnosis: A Multi-Method Approach* (RIT M.S. Computer Science graduate research, 2015) — ensemble and deep learning methods for clinical classification on high-dimensional biomedical data.
- *Predicting Hospital Readmission Rates: A Multi-Method ML Analysis* (DSCI-633 Project Report, 2025) — ensemble and regularized regression methods for clinical risk stratification under distribution shift.
- *BIO614-FinalProjectProposal* (Overleaf manuscript, 2026) — RNA secondary structure prediction with thermodynamic deep learning enhancements, reproducible benchmarking, and biological interpretation.
- *Equitable Bioinformatics: Enhancing Diagnostic Decision-Making through RNA and Biomarker Data* (ISTE-780 Project Report, 2025) — fairness-aware ML for genomic diagnostic algorithms with SHAP-based bias detection and statistical significance testing.
- *Differential Gene Expression in Murine DRG Neurons Following Sciatic Nerve Injury: An NGS Reanalysis* (BIOL550 Computational Biology Project, 2026) — reproducible bulk RNA-seq pipeline using DESeq2, QC validation, and pathway-level biological interpretation of nociceptor gene expression.
- *Fairness-Aware Bandits for Clinical Decision Systems* (DSCI-601 Project Proposal, 2025) — bandit learning with equity constraints applied to high-stakes clinical diagnostic decision-making, addressing demographic disparity in sequential clinical recommendation workflows.

Awards & Recognition

- NSF Robert Noyce Teacher Scholarship (2024–2026) — full funding for M.S. in Teaching Computer Science K–12.
- MS International Scholarship (2012–2015) — MESCYT & RIT, awarded for academic excellence.
- Honorable Mention Technology Award (2012) — NYS STEP/CSTEP, 2nd place capstone project.
- IEEE Alumni Member (2009–Present) — Member No. 90613000.

Technical Competencies

Programming: Python, R, Java, C++, C#, SQL, MATLAB, JavaScript, Bash

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost; bandit/RL (UCB, contextual bandits, Thompson sampling); fairness-aware ML; model evaluation and uncertainty quantification; SHAP analysis

Data and Statistics: pandas, NumPy, SciPy, statistical modeling, hypothesis testing, reproducible benchmarking

Infrastructure: Git/GitHub, Docker, Linux, GCP, AWS, LaTeX

Selected Fit for KTH Machine Learning Position

- Strong theoretical and applied ML background spanning supervised, sequential decision-making, and fairness-aware methods, with emphasis on rigorous evaluation methodology.
- Hands-on experience designing and running controlled ML experiments with statistical validation — directly applicable to research that advances ML methodology.
- Demonstrated commitment to reproducibility: all research projects include structured evaluation pipelines, logging, and statistical-significance reporting.
- Deep interest in understanding when and why ML methods work or fail, with particular focus on robustness, equity, and reliability in high-stakes applications.